

Morena Porzio

Curriculum Vitae

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Education and Formation

- 2025-now **Postdoctoral Fellow at the Department of Mathematics**, *University of Toronto*, 40 St. George Street; Toronto, ON.
- 2020-2025 **PhD Student in Mathematics**, *Columbia University, GSAS*, New York, NY.
- 2022-2024 **Master of Philosophy in Mathematics**, *Columbia University, GSAS*, New York, NY.
- 2020-2022 **Master of Arts in Mathematics**, *Columbia University, GSAS*, New York, NY, GPA: 4.1.
- 2015-2020 **Master's & Bachelor's Degree in Mathematics**, *Scuola Normale Superiore, Honors College*, Pisa, Italy, 100/100 cum Laude, GPA: 4.0.
- 2018-2020 **Master's Degree in Mathematics**, *University of Pisa*, Pisa, Italy, 110/110 cum Laude, GPA: 4.0.
- 2015-2018 **Bachelor's Degree in Mathematics**, *University of Pisa*, Pisa, Italy, 110/110 cum Laude, GPA: 4.0.
- 2010-2015 **High School Diploma**, *Liceo Scientifico F. Vercelli*, Asti, Italy, 100/100 cum Laude.

Publications and Preprints

- 2025 **On Modified Diagonal Cycles and the Beauville Decomposition of the Ceresa Cycle**, *L. Lagarde, M. Moakher, M. Porzio, J. Rawson, F. Trejos Suárez*, Preprint: <https://arxiv.org/pdf/2510.07416>.
- 2025 **Isotopy invariance and stratified $\mathbb{E}_2$ -structure of the Ran Grassmannian**, *G. Nocera, M. Porzio*, Preprint: <https://arxiv.org/pdf/2509.06222>.
- 2025 **Density of algebraic points on products of curves**, *J. Berg, Y. Fu, E. Gazaki, M. Porzio, J. Rawson, I. Vogt*, Preprint: <https://arxiv.org/pdf/2507.00860v1>.
- 2025 **The Arithmetic of Del Pezzo Surfaces and Hilbert Schemes of Points**, Preprint: <https://academiccommons.columbia.edu/doi/10.7916/1hz7-fc26>.
This is my PhD Thesis done at Columbia University: my advisor was Aise Johan de Jong.
- 2024 **On the Stable Birationality type of Hilbert schemes of points**, *M. Porzio*, Preprint <https://arxiv.org/pdf/2408.07593>.
- 2022 **The Torsion-Free Rank of $Ext^1(A, \mathbb{Z})$ and Whitehead's Problem**, *M. Porzio*, *The Graduate Journal of Mathematics*, Volume 7, Issue 1 (2022), 10-16 <https://gradmath.org/wp-content/uploads/2022/09/GJM2022-Porzio.pdf>.
For a longer version (Bachelor Thesis), see http://www.logicatorino.altervista.org/matteo_viale/Porzio.pdf
- 2022 **An Estimation of Hailstone Radii Using the Collatz Conjecture**, *R. Gupta, M. Porzio*, *Journal of Student Research (JSR)*, Vol. 11 No. 2 (2022) <https://doi.org/10.47611/jsrhs.v11i2.3638>.
Mentorship through Lumiere education, Premium Research & Publication Program.
- 2020 **Picard Groups of Stacks of Elliptic and Hyperelliptic Curves**, *M. Porzio*, Preprint: <https://etd.adm.unipi.it/t/etd-09102020-102635/>.
This is my Master Thesis done at University of Pisa: my advisor was Angelo Vistoli.

Ongoing Projects

- **Generators and Relations of the Tautological Subring of a Jacobian**, *joint work with L. Lagarde, M. Moakher, J. Rawson, and F. Trejos.*
This work extends the work of Polishchuk on the sl_2 -action on $\mathrm{CH}(J)_{\mathbb{Q}}$ and defines an *unstable* version of tautological rings for C^n .
- **Classification of DP3's over finite fields**, *joint work with J. Desjardins, A. Knecht, J. Schneider, M. Trip.*
Using Hirschfeld's theorems on the configuration of Eckardt points on DP3's, we give a complete classification via isomorphism classes of DP3's over finite fields.
- **Project on full DP2's over characteristic 2**, *joint work with J. Desjardins, A. Knecht, J. Schneider, M. Trip.*
The aim of this project is to complete the classification of *full* DP2's over finite fields.
- **Study of the Albanese map from Symmetric powers**, *joint work with N. Chen, A. Fernandez Herrero, T. Krämer.*
- **Study the Density Degree Set in Families**, *joint work with L. Beneish, N. Chen, S. Memarian.*
- **Intermediate Jacobians Obstruction and Conic bundles with singular Discriminant locus**, *joint work with S. Abe, E. Feng, D. Genlik, Y. Huang, L. Ji, F. Okamura, S. Sharma.*

Grants and Awards

- 2026 **Nomination for Full Membership in *Sigma Xi, The Scientific Research Honor Society***, Nominated by J. L. Vernon, and A. A. Thomas, <https://www.sigmaxi.org/>.
- 2023/2024 **Winner of PhD Candidate Recognition Award for Excellence in Teaching**, *Math Department of Columbia University*, New York, NY.
- 2020-2026 **Winner of Fellowship Package as a Dean's Fellow and doctoral student at the Math Department of Columbia University**, New York, NY.
- 2022-2024 **Sponsored Member of the Association for Women in Mathematics, AWM**, <https://awm-math.org/>.
- 2019 **Winner of the AILA Award 3+2 2019**, *Associazione Italiana di Logica e sue Applicazioni (Italian Association of Logic and its Applications)*.
<http://www.ailalogica.it/2019/06/03/vincitori-premio-aila-32-anno-2018-19/>
- 2015-2020 **Winner of Full Scholarship with Grant at Scuola Normale Superiore, SNS**, Pisa, Italy.
- 2015 **Winner of INdAM Scholarship**, *National Institution of High Mathematics "Francesco Severi"*, https://www.altamatematica.it/sites/default/files/graduatoria40-2015-2016_0.pdf.
- 2015 **National Gold Medal, Italian Mathematical Olympiads (preIMO)**, Cesenatico, Italy.
Scored as the ninth Italian high-school student, <http://olimpiadi.dm.unibo.it/archivio/medagliere/>.
- 2015 **International Bronze Medal, European Girls' Mathematical Olympiad (EGMO)**, Minsk, Belarus.
Scored in the top 50 worldwide participants, <https://www.egmo.org/egmos/egmo4/scoreboard/>.
- 2015 **National Bronze Medal, Italian Physics Olympiads (preIPhO)**, Senigallia, Italy.
Scored in the top 40 Italian high-school students <https://www.olifis.it/>

Teaching and Mentoring

- Year 24-25 **Math Grad Representative**, *Columbia University*, New York.
- Spring '25, **TA of MATHUN2030**, *Ordinary Differential Equation*, *Columbia University*, New York.
- Spring '24, **TA of MATHGR6261**, *Commutative Algebra*, *Columbia University*, New York.
- Fall '22, **Graduate Student Editor at** *Columbia Journal of Undergraduate Mathematics*,
Spring '22, <https://journals.library.columbia.edu/index.php/cjum>.
Fall '21
- Fall '24, '23 **TA of MATHGR6262**, *Arithmetic and Algebraic Geometry*, *Columbia University*, New York.
- Summer '23 **Instructor of MATHUN2030**, *Ordinary Differential Equations*, *Columbia University*, New York.
- Summers **Mentor at Lumiere Education**, Premium Research and Publication Program, <https://www.lumiere-education.com/>, Online.
'23, '22
- Spring '23 **TA of MATHGU4043**, *Algebraic Number Theory*, *Columbia University*, New York.
- From '21 to **Administrator of the Putnam Exam**, *Columbia University*, New York, NY.
'24
- Spring '21 **Young Mentor in THE STEM factor project**, *Global Shapers* <https://www.globalshapers.org/>, Florence, Italy.
Cycle of interviews and Q&A sessions, whose main goal is to address gender issues among the STEM world.
- Fall '18 **TA of the master course Algebra**, *University of Pisa*, Pisa, Italy.
- Apr '18 **Staff member at EGMO 2018**, <http://www.egmo2018.org/>, Florence, Italy.
Staff member and invigilator at the European Girls' Mathematical Olympiad, an international math competition.
- Springs '16, **Trainer at Italian EGMO training**, *Staff Member – Problems proposer and grader*.
'17, '18
- Winter '17 **Co-instructor at Math Olympiads Internship**, *UMI – Unione Matematica Italiana (Italian Mathematics Union)*, Massa, Italy.
- Spring '16 **Staff Member of the National Mathematical Olympiads**, *UMI*, Cesenatico, Italy.
Corrector of the problems proposed and a staff member at the national mathematical competition.

Invited Talks at Seminars

- Feb 11 2026 **U of T number theory seminar**, "Measuring" the Vanishing of Ceresa cycles via Modified diagonals, <https://alex-youcis.github.io/nt-seminar.html>, *University of Toronto*, Toronto, Canada.
- Dec 5 2025 **Algebra Seminar**, *The Cassels–Swinnerton-Dyer Conjecture and Stable Birationality of Symmetric Powers of X*, https://math.unt.edu/research/algebra_seminar.html, *University of North Texas*, Denton, US.
- Nov 5 2025 **AG Seminar**, *The Cassels–Swinnerton-Dyer Conjecture and Stable Birationality of Symmetric Powers of X*, <https://sites.google.com/view/agseminar/home>, *University of Toronto*, Toronto, Canada.
- Jul 11 2025 **Summer Research Institute in Algebraic Geometry, Bootcamp, Poster Session**, *On the Stable Birationality of Hilbert schemes of points on surfaces*, <https://sites.google.com/view/agbootcamp2025/schedule?authuser=0>, *Colorado State University*, Fort Collins, US.

- Jun 24 2025 **Algebraic Points on Curves, Lightning Talk - On the stable birationality of Hilbert schemes of points on surfaces**, https://icerm.brown.edu/program/topical_workshop/tw-25-apc, ICERM, Providence.
- May 17 2025 **2025 AWM Research Symposium, Higher Beauville Components of Ceresa Cycles and Modified Diagonals**, <https://awm-math.org/meetings/awm-research-symposium/>, University of Wisconsin-Madison, Madison.
- Apr 4 2025 **UVA Number Theory Seminar, On the stable birationality of Hilbert schemes of points on surfaces**, <https://math.virginia.edu/seminars/ntsem/>, University of Virginia, Charlottesville.
- Dec 3 2024 **Utah Algebraic Geometry Seminar, On the stable birationality of Hilbert schemes of points on surfaces**, <https://www.math.utah.edu/agseminar/>, University of Utah, Salt Lake City.
- Nov 19 2024 **Emory Algebra and Number Theory Seminar, On the stable birationality of Hilbert schemes of points on surfaces**, <https://sarangop1728.github.io/eants>, Emory University, Atlanta, GA.
- Nov 11 2024 **Association for Women in Mathematics, On the density of algebraic solutions on products of curves**, <https://www.math.columbia.edu/diversity/awm/>, Columbia University.
- Nov 4 2024 **Poster Session at Instructional Workshop on Rational Points, On the stable birationality of Hilbert schemes of points on surfaces**, University of Groningen <https://www.rationalpoints.nl/events-2/instructional-workshop-on-rational-points/>.
- Nov 4 2024 **Seminar on Arithmetic Geometry and Algebraic Groups, On the stable birationality of Hilbert schemes of points on surfaces**, <https://seminaragag.github.io/>, Online.
- Oct 30 2024 **University of Michigan algebraic geometry seminar, On the stable birationality of Hilbert schemes of points on surfaces**, <https://events.umich.edu/event/126925>, University of Michigan, Ann Arbor.
- Oct 24 2024 **Tufts Algebra, Geometry, and Number Theory Seminar, On the stable birationality of Hilbert schemes of points on surfaces**, <https://sites.google.com/view/tuftsagents/home>, Tufts University, Boston.
- Jun 28 2024 **WIAG 2** <https://sites.google.com/view/wiag2024/>, Project Group presentation on Classification of del Pezzo surfaces, IAS, Institute for Advanced Study, Princeton.
- Oct 1 2024 **Poster Session at Workshop on Algebraic Cycles, On the stable birationality of Hilbert schemes of points on surfaces**, <https://sites.google.com/view/workshop-on-algebraic-cycles>, Leibniz Universität, Hannover.
- Apr 10 2024 **Columbia Undergraduate Math Society, Spring 2024, Riemann–Roch Theorem for Graphs and Specialization of Divisors from Curves**, <https://math.columbia.edu/~ums/>, Columbia University, NY.
- Mar 22 2024 **Degree d -points on Algebraic Surfaces - AIM Workshop** <https://aimath.org/workshops/upcoming/degreesurface/>, Stable Birationality Types of Hilbert schemes of points on Surfaces, Caltech University, Pasadena, CA.
- Mar 6 2024 **Arizona Winter School 2024: Abelian Varieties, Spring 2024**, <https://swc-math.github.io/aws/2024/index.html>, Project Group presentation on Ceresa's cycles, University of Arizona, Tucson, AZ.
- Apr 19 2023 **Columbia Undergraduate Math Society, Spring 2023, Obstructions to the existence of rational points on algebraic varieties**, <https://www.math.columbia.edu/~ums/2023spring.html>, Columbia University, NY.

Organized Learning Seminars

- Spring 2025 **Talk till you are stuck, AG Research Seminar**, <https://www.math.columbia.edu/~morenaporzio/TTURTseminar.html>.
- Spring 2024 **Elliptic Curves and Abelian Varieties**, <https://www.math.columbia.edu/~morenaporzio/PAWSseminar.html>.
- Spring 2022 **Motives, Standard Conjectures and Weil Cohomology Theories**, <https://math.columbia.edu/~morenaporzio/Motives&WCT.html>.

Talks at Learning Seminars

- Apr 19 2024 **Spring 2024 AG learning seminar: Decomposition of the diagonal, Nilpotence conjecture and Kimura's theorem**, https://yoonjookim.github.io/seminars/spring_2024.html, Columbia University.
- Apr 12 2023 **Preprint Seminar, Spring 2023, Smooth subvarieties of Jacobians**, <https://www.math.columbia.edu/~tpad/preprintS2023.html>, Columbia University.
- Feb 24 2023 **Spring 2023 Informal AG Seminar on MMP, Minimal Model Program in dimension 2**, <https://www.math.columbia.edu/~andresfh/sp23agseminar.html>, Columbia University.
- Feb 16 2023 **Milnor-Witt motives seminar 2023, Milnor-Witt K-theory**, <https://math.columbia.edu/~magenroy/MilnorWittMotivesSeminar.html>, Columbia University.
- Nov 16 2022 **Rigid Analytic Geometry and Perfectoid Spaces, Generalities on Constructible Sets and Spectral Spaces**, <https://www.math.columbia.edu/~scheffelin/perfectoid>, Columbia University.
- Nov 9 2022 **Non-abelian Hodge correspondence and $P=W$ conjecture, Perverse sheaves and Lefschetz hyperplane theorem**, <http://math.columbia.edu/~anabasheva/pw.html>, Columbia University.
- Oct 14 2022 **Motivic homotopy theory learning seminar, Basics of the unstable motivic homotopy category**, <http://math.columbia.edu/~magenroy/motivicseminar.html>, Columbia University.
- Nov 24 2021 **SNS Dissertation Talk, Surjectivity of Brauer Map for surfaces: positives results and counterexample**, Zoom – Scuola Normale Superiore.
- Oct 29 2021, **Deformation Theory Seminar, Application of deformation theory to contracting**
Nov 5 2021 **curves functor $\mathcal{M}_{g,n+1} \rightarrow \mathcal{M}_{g,n}$** , <https://www.math.columbia.edu/~dejong/seminar/seminar-deformation-theory.html>, Columbia University.
- Apr 27 2021, **Derived Algebraic Geometry Learning Seminar - Formal moduli problems**,
May 5 2021, **Ring Spectra and Generalized Cohomology Theory**, <https://uz.sns.it/~Wilhelm795/FMP2021.html>, Zoom – Scuola Normale Superiore.
- May 11 2021 **FMP2021**, Zoom – Scuola Normale Superiore.
- Mar 26 2021 **Intersection Theory Learning Seminar, Grothendieck-Riemann-Roch**, <https://www.math.columbia.edu/~plei/s21-INT.html>, Columbia University.
- Oct 30 2020, **Geometric Invariant Theory Learning Seminar, GIT and Coarse Moduli Problems**,
Nov 6 2020 <https://www.math.columbia.edu/~plei/f20-GIT.html>, Columbia University.

Schools and Conferences

- July 14-31 **Summer Research Institute in Algebraic Geometry, Colorado State University, Fort**
2025 **Collins**, <https://sites.google.com/view/2025summerinstitute/home>.
- July 8-12 **Summer Research Institute in Algebraic Geometry, Bootcamp, Colorado**
2025 **State University, Fort Collins**, <https://sites.google.com/view/agbootcamp2025/home?authuser=0>.

- June 23-27 **ICERM - Algebraic Points on Curves**, *Brown University*, Providence, https://icerm.2025.brown.edu/program/topical_workshop/tw-25-apc.
- May 16-18 **2025 AWM Research Symposium**, *University of Wisconsin-Madison*, Madison, <https://awm-math.org/meetings/awm-research-symposium/>.
- Nov 4-8 **Instructional workshop on rational points**, *University of Groningen*, Netherlands, <https://www.rationalpoints.nl/events-2/instructional-workshop-on-rational-points/>.
- Sept30-Oct2 **Workshop on Algebraic Cycles**, *Leibniz Universität*, Hannover, <https://sites.google.com/view/workshop-on-algebraic-cycles>.
- June 24-28 **Collaborative Workshop in Algebraic Geometry**, *IAS - Institute for Advanced Studies*, Princeton, <https://www.ias.edu/math/events/collaborative-workshop-algebraic-geometry>.
- May 6-10 **The Geometry of Hilbert Schemes of Points**, *CIRM*, Levice Terme (Italy), <https://sites.google.com/view/ghisp2024/home>.
- Apr 22-26 **On the occasion of Hélène Esnault's 70th birthday**, *IHES - Centre de conférence Marilyn et James Simons*, Bures-sur-Yvette (Paris), <https://indico.math.cnrs.fr/event/11114/>.
- Mar 18-22 **Degree d points on algebraic surfaces**, **AIM Workshop**, *Caltech University*, Pasadena (CA), <https://aimath.org/workshops/upcoming/degreesurface/>.
- Mar 2-6 **Arizona Winter School 2024: Abelian Varieties**, *University of Arizona*, Tucson (AZ), <https://swc-math.github.io/aws/2024/index.html>.
- Dec 7-9 **Monodromy and Its Applications**, **Conference on the Occasion of Nicholas Katz' 80th Birthday**, *Princeton University*, Princeton, <https://sites.google.com/princeton.edu/katz80>.
- June 19-23 **Conference on 100 Years of Noetherian Rings**, *IAS - Institute for Advanced Studies*, Princeton, <https://www.ias.edu/math/events/conference-100-years-noetherian-rings>.
- June 12-16 **Arithmetic, Birational Geometry, and Moduli Spaces**, *Brown University*, Providence, <https://sites.google.com/view/abgms2023/>.
- Nov 18-20 **AGNES Fall 2022 (Algebraic Geometry Northeastern Series)**, *University of Massachusetts*, Amherst, <https://sites.google.com/site/agneshomepage/amherst-2022?authuser=0>.
- Oct7, Nov4 **ENYGMMA (Empowering New York Gender Minority Mathematicians)**, *CUNY, Columbia University, Stony Brook University*, New York City, <https://sites.google.com/stonybrook.edu/enygmma/home?authuser=0>.
- Sept 12-16 **Summer school "Motives in Ratisbona"**, *University of Regensburg*, Regensburg, <https://www.uni-regensburg.de/mathematics/motives2022/startseite/index.html>.
- May 31-4(Jun) **O-minimality and foliations**, *CIRM (Centre International de Rencontres Mathématiques)*, Virtual Workshop, <https://conferences.cirm-math.fr/2682.html>.
- 2021
- Jun 8-20 **Algebraic and Tropical Online Meetings (ATOM)**, Virtual School, <https://researchseminars.org/seminar/ATOM>.
- 2020
- Aug 26-1(Sept) **Summer School of Logic**, *AILA (Italian Association of Logic and its Applications)/SILFS (Italian Society of Logic and Philosophy of Science)*, Gargnano, Italy, <http://sel.di.unimi.it/>.
- 2019
- Nov 27-30 **Young Perspectives in Deformation Theory**, *Politecnico of Turin*, Turin, <https://sites.google.com/site/ypdt2018/>.
- 2018